Two-Tier Labor Markets and Durable Consumption: The Role of Risk Heterogeneity

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Abstract

This paper examines the impact of two-tier labor markets on business cycle volatility through durable consumption choices. To analyze two-tier labor markets and risk heterogeneity, I focus on labor contracts. Using Italian panel data, I observe that during recessions, workers on fixed-term contracts reduce their durable consumption significantly more than their permanent counterparts. I use an incomplete market model of durable and non-durable consumption with a dual labor market, where durable consumption is subject to a non-convex adjustment cost. In this framework, I highlight the heterogeneous effects of changes in upside versus downside income risk on the magnitude and persistence of durable consumption adjustments. I also show that harmonizing contract types offers a promising tool for stabilizing the business cycle.

Keywords: Durable Consumption, Labor contracts, Great Recession

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1 Introduction

In many labor markets, employed workers are segmented into different tiers, such as open-ended versus fixed-term contracts, payroll versus staffed positions, year-round versus seasonal or interim employment, and formal versus informal workers. These tiers are associated with different levels and volatility of income and unemployment risk. What is the impact of such risk heterogeneity over the business cycle? How do labor market institutions interact with aggregate stabilization?

I focus on two-type contract labor markets, divided between high-security permanent contract workers and low-security fixed-term contract workers. In 2022, fixed-term employment — defined as workers with a predetermined termination date — accounted for an average of 12% of the workforce in OECD countries. I use Italy as a case study and examine how this division impacts the business cycle through consumption and, in particular, durable consumption. This channel is important because durable consumption is one of the main drivers of business cycle volatility.[1] I show that in Italy, during the Great Recession, fixed-term contract holders reduced their durable purchases significantly more than their permanent counterparts, and that labor contract heterogeneity plays a role in business cycle volatility.

This difference can be explained by the distinct nature of risk faced by the two groups of workers. Over the Great Recession, the main driver of permanent contract holders' consumption decisions was an increase in their downside income risk. In contrast, fixed-term contract holders primarily responded to a reduction in their upside income risk. The tail of the income distribution from which the change in risk originates significantly affects the magnitude and persistence of durable consumption contractions. It follows that this result can also be applied to labor markets where the two-tier division is not formalized through distinct contract types but instead occurs more informally, as in the case of the U.S., provided that workers face changes in risk at different ends of the income distribution.

I gather motivational evidence from the Italian Survey of Households Income and Wealth (SHIW) and use car purchases to measure households' durable consumption. I observe a sharp and uneven drop in car purchases during the Great Recession. Consistent with Attanasio et al.'s (2022) results for the US, Italian households reduced car purchases on both the extensive and intensive margins. The extensive margin

[1] Broad durable goods accounted for more than half of the overall drop in real U.S. GDP between 2007 and 2009 (Berger and Vavra, 2015). For further evidence of the excess volatility of durables, see, among others, Atalay (2017), Bernanke (1985), Dupor et al. (2020), Kydland and Prescott (1992), Leamer (2007), and Mankiw (1982).

refers to the share of households buying a car, while the intensive margin is the amount spent, conditional on purchasing. A novel finding of this paper is that the drop in car consumption was significantly larger among households whose members are employed on permanent contracts than among households whose members are employed with fixed-term or temporary contracts.[2] For brevity, I will refer to these groups as permanent households and fixed-term households in the remainder of the paper. During the Great Recession, the share of fixed-term households purchasing a car dropped by more than 40%, twice the decline observed among permanent households. On the intensive margin, fixed-term households reduced spending by 17%, compared to 12% for permanent households.

To rationalize these facts, I build a model of household consumption and saving behavior. The model is an incomplete market model in which households consume durable and non-durable goods. Crucially, durable purchases are subject to a non-convex adjustment cost *a la* Grossman and Laroque (1990). Following the empirical specification, I model two types of contract, each associated with a particular unemployment risk and income process. The policy functions derived from the model show that households update their durables according to a trigger-target (S,s) rule.[3] As durable investments are partially irreversible, households wish to limit the frequency of such purchases. This result is consistent with the empirical literature documenting lumpy durable consumption patterns. The model is disciplined with the SHIW data, focusing on the distinct income and unemployment risk processes associated with each contract type. I use the model to disentangle the driving forces behind the contraction in durable consumption across contract types and adjustment margins. A placebo experiment separates the effects of changes in households' income risk from realized income losses.

For permanent households, changes in income risk during the Great Recession accounted for more than 40% of the baseline decline in the extensive margin of car purchases, but none of the decline in the intensive margin.[4] The income distribution of permanent households is negatively skewed. During the Great Recession, their risk of unemployment or of being downgraded to a fixed-term contract increased, raising

[2] I focus on households in which all employed members have the same type of employment contract or households with a single income earner. This restriction ensures a clean classification into either the permanent or fixed-term contract group.

[3] (S,s) dynamics are studied by Attanasio (2000), Bar Ilan and Blinder (1992), Caballero (1993), Caballero and Engel (1999), Carroll and Dunn (1997), Eberly (1994) and Foote et al. (2000), among others.

[4] This is in line with Bertola et al. (2005), who show that the two margins of durable consumption are driven by different economic factors.

income variance. Because of non-convex adjustment costs, purchasing a car represents a commitment that permanent households are less willing to undertake under heightened downside income risk, leading them to postpone durable purchases until uncertainty subsides.[5] The increase in downside income risk drives the car adjustment response of permanent households during the recession. I label the associated strategy as "Wait-to-See".

The mechanism at play is markedly different for fixed-term households. For them, the change in income risk during the Great Recession accounted for changes along both the extensive and intensive margins of car purchases. Unlike permanent households, fixed-term households have a positively skewed income distribution, with large income changes mainly coming from upgrades to permanent contracts. However, during the Great Recession, the probability of such upgrades dropped significantly, reducing their income variance by nearly 30%. With greater persistence in their high-risk, low-income employment status, fixed-term households rebalance their portfolios away from durables, since the partial irreversibility of durables makes them a poor hedge against labor income fluctuations. Consequently, they lower their optimal target stock of durables. Although these households want to downsize their cars, high adjustment costs make this difficult. Instead, they rely on depreciation to achieve downsizing while avoiding transaction costs. The decline in upside income risk drives fixed-term households' car response, and I label the strategy as "Wait-to-Rebalance." The literature has focused on how a rise in income variance can lead households to become stuck within the inaction band of durable consumption. Here, I show that when the change in risk arises from a collapse in upside income risk, a decrease in income variance can also keep households within the durable consumption inaction band.

Does distinguishing between "Wait-to-See" and "Wait-to-Rebalance" matter? First, the driver of the drop in durable goods purchases — specifically upside versus downside income risk — significantly impacts the duration of the contraction. The "Wait-to-See" strategy triggers a sharp but short-lived decline in car purchases, while "Wait-to-Rebalance", driven by the slow process of depreciation, results in a very persistent decline throughout the recession.

Second, distinguishing between upside and downside income risk is crucial when evaluating public policy. I use the model as a laboratory to examine the impact of a car subsidy equivalent to 5% of car expenditures in 2009.[6] The policy has limited

[5] In line with Bernanke (1983) and Dixit and Pindyck (1994).

[6] Related to the literature on how durable consumption responds to cash-for-clunkers programs

ability to stimulate durable consumption among households experiencing a decline in upside income risk. It is, *de facto*, partly transformed into a subsidy for downgrading existing cars.

Finally, I show that harmonizing contract types offers a promising tool for stabilizing the business cycle. Using the model, I compare the depth of the Great Recession in Italy's dual-contract labor market with simulations of a single-contract labor market. I assume the average level of risk remains unchanged but is uniformly shared by all workers, resulting in a single contract that bears the average risk. I explore several scenarios of how unemployment volatility responds to contract harmonization. First, if unemployment volatility does not respond to contract harmonization, a single contract reduces the decline in car consumption by 13% on the intensive margin, but increases the contraction on the extensive margin by 10%. This ambiguous effect is driven by two opposing forces: on the one hand, the strong and persistent consumption contraction of fixed-term workers is muted, but on the other hand, most of the risk in a two-contract system is borne by a small segment of the workforce (the fixed-term contract holders), which accounts for a small share of aggregate consumption. Second, following Costain et al. (2010) and Bentolila et al. (2012), I consider that contract harmonization can reduce unemployment volatility. The stabilization effect of contract harmonization is more pronounced. I find that a unified labor contract could have reduced the contraction in durable consumption during the Great Recession by up to 13% on the extensive margin and 5% on the intensive margin.

Related Literature. The contribution of this paper is to study the impact of risk heterogeneity on business cycle volatility through durable consumption, particularly by disentangling the effects of changes in upside versus downside income risk. I study propensities to consume out of (heterogeneous) changes in risk.

The paper fits in the literature on lumpy durable consumption in macro models. The closest work is Harmenberg and Oberg (2021), who estimate the consumption response to labor market shocks using SHIW data, finding that income uncertainty from higher unemployment risk drives the decline in durable goods spending. Berger and Vavra (2015) show that non-convex adjustment frictions lead to state-dependent policy responses, while McKay and Wieland (2021) discuss the inter-temporal trade-off for monetary policy in stabilizing durable demand. These papers focus on *ex-post*

(Adda and Cooper, 2000; Berger et al., 2024; Mian and Sufi, 2012) and fiscal policy (Bachmann et al., 2024; Beraja and Zorzi, 2024; Fuster et al., 2021; Orchard et al., 2024; Parker et al., 2013; Parodi, 2024).

heterogeneity from transitory income shocks. In contrast, contract type represents persistent risk, lying between *full ex-ante* and *ex-post only* heterogeneity. The impact of this persistent heterogeneity has not been explored. Unlike these studies, I also separate the intensive and extensive margins of durable consumption to disentangle the driving forces for each contract type. In this respect, this paper is related to Attanasio et al. (2022), who first study both adjustment margins in a life-cycle model.

This work also relates to the literature on income and consumption dynamics (see, for example, Almuzara et al., 2025; Blundell and Preston, 1998; Guvenen et al., 2014, 2021; Kaplan and Violante, 2014), as well as the impact of uncertainty on household and firm decisions (see Bayer et al., 2019; Bloom, 2009; Bloom et al., 2018; Justiniano and Primiceri, 2008; and Storesletten, et al., 2004, among others). Here, I focus on upside and downside income risk, showing that this dimension influences the magnitude and persistence of durable consumption contractions during a recession.

Finally, this paper contributes to the rich literature on dual employment protection laws (see Bentolila et al., 2019 for a review), which primarily focuses on employment, wages, or productivity. In contrast, this paper examines the impact of labor market duality on the business cycle through consumption. The relationship between labor market structure and consumption is little explored. [7]

2 Motivating Evidence

I use data from the Survey on Household Income and Wealth (SHIW) conducted by the Bank of Italy. The main strength of the SHIW dataset is that, alongside many demographic and job-related characteristics, it provides detailed information on households' income, wealth, durable goods expenditures, and non-durable consumption. I use the SHIW waves between 2000 and 2014 to compare changes in employment status, income, and consumption between workers employed with permanent contracts and those with fixed-term or temporary contracts. When variables are reported at the household level (e.g., consumption), I compare households whose members are employed with permanent contracts (referred to as permanent households for convenience) and those whose members are employed with fixed-term or temporary contracts (referred to as fixed-term households). I focus on households in which all employed members have the same type of employment contract or house-

[7] Another contribution in this area is Dekle and Nakamura (2024) who study how dual labor market affects the consumption response to monetary policy in Japan.

holds with a single income earner. This restriction ensures a clean classification into either the permanent or fixed-term contract group. The data frequency is biannual. A further presentation of the dataset, details on variable definitions, sample selection, and summary statistics can be found in Appendix A.

Table 1 displays changes in income, non-durable consumption, and detailed car purchases for each group during the Great Recession. The Great Recession refers to the 2008-2014 period, which includes both the 2008-2009 recession and the recessionary episode of 2011-2013, sometimes referred to as the sovereign debt crisis. One aim of this paper is to study the extensive and intensive margins of durable consumption separately. Studying changes in the intensive margin across a basket of goods can be misleading.[8] Therefore, I restrict the analysis to a single durable good: cars. I choose cars because, abstracting from houses as done in this paper, they represent the largest category of durable goods purchased by households (between 2002 and 2014, cars alone accounted for roughly 30% of all durable purchases). In the remainder of the paper, I use "cars" and "durables" interchangeably.

Table 1: Consumption and Income Response over the Great Recession

	Cars ext. margin (%)		**Cars int. margin (€)**		**Non-durable cons. (€)**		**Income (€)**	
	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.
Boom	15.76	12.53	6,869	5,458	15,549	11,180	35,005	18,807
Recession	12.42	6.96	6,026	4,507	15,192	11,175	32,778	16,859
Change	**-0.21**	**-0.44**	**-0.12**	**-0.17**	**-0.02**	**0.00**	**-0.06**	**-0.10**

Notes: Perm. stands for permanent contract and F.t. for fixed-term or temporary contract. Boom is 2002-2006, Recession is 2008-2014. cons. stands for consumption, ext. for extensive and int. for intensive. Income is income from labor and transfers. Top and bottom 1% are winsorised. Households sampling weights are used.

Table 1 highlights a stark and unevenly distributed drop in car purchases during the Great Recession. First, in line with the findings of Attanasio et al. (2022), households significantly reduced their car purchases across both adjustment margins. Second, and this is a novel finding of this paper, the drop in car consumption was significantly larger for fixed-term households. Over the recession, the share of fixed-term households purchasing a car in a given year dropped by more than 40%, twice as much as for permanent households. On the intensive margin, fixed-term households reduced their spending by 17%, compared to 12% for their permanent counterparts. Finally, both groups roughly maintained their non-durable purchases during the

[8] If agents buy a car in one year and a sofa in the following year, it does not mean they decreased their intensive margin of durable consumption. The drop in purchase value simply reflects different types of investment.

recession. This holds despite a drop in mean income from labor and transfers of 6% and 10% for permanent and fixed-term households, respectively.

Beyond income and consumption patterns, different types of contracts also face heterogeneous levels and fluctuations in employment risk. Table 2 shows the transition frequencies of workers' employment status before and during the Great Recession. During the expansionary period, workers employed with fixed-term contracts had a 49% probability of being employed with a permanent contract in the next wave of the survey (i.e., two years later), and a 15% probability of becoming unemployed over the same time horizon. During the Great Recession, the two-year probability of upgrading to a permanent contract dropped to 33%, while the unemployment risk increased to 20%. Permanent contract holders cannot upgrade to a more stable contract type, but they can either downgrade to a fixed-term contract or become unemployed. Over a two-year horizon, these events occurred with probabilities of 4% and 3% during the boom, and 6% and 5% during the recession.

Table 2: Probability to Change Employment Status (Two-year Horizon)

	Upgrade to Perm.		**Downgrade to F.t.**		**Downgrade to Unemployed**	
	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.
Boom	-	0.49	0.04	-	0.03	0.15
Recession	-	0.33	0.06	-	0.05	0.20

Notes: Perm. stands for permanent contract and F.t. for fixed-term or temporary contract. Boom is 2002-2006 and Recession is 2008-2014.

3 The Model

I formulate a model of household consumption and saving behavior in which the labor market is segmented into two types of contracts, each associated with a distinct unemployment risk and income process. The model is a partial equilibrium incomplete markets model in which households consume durable and non-durable goods. Durable purchases are subject to a market friction. This household choice setting is close to Berger and Vavra (2015) or Harmenberg and Oberg (2021).

3.1 The Household Problem

The economy is populated by a continuum of *ex-ante* identical households of measure one, indexed by $i \in [0, 1]$. Households are infinitely lived, time is discrete, and each period corresponds to a quarter. Households supply labor inelastically.

Preferences

Households derive utility from their non-durable consumption (c_t) and their stock of durable goods (D_t). They discount the future at rate β. The value function of household i can be written as:

$$V_i = E_0 \max_{\{c_{it}, D_{it}\}} \sum_{t=0}^{\infty} \beta^t u(c_{it}, D_{it})$$

$$with \quad u(c_{it}, D_{it}) = \frac{\left[c_{it}^{\alpha} D_{it}^{(1-\alpha)}\right]^{(1-\sigma)}}{1-\sigma}$$

where α is the weight of non-durable consumption in the utility function and σ is the coefficient of relative risk aversion.[9]

Idiosyncratic risk - employment risk

Households face idiosyncratic employment risk. In any given period, a household can be employed under a permanent contract, employed under a fixed-term contract, or unemployed. Households with fixed-term contracts face a higher probability of becoming unemployed than those with permanent contracts. Transitions between the three employment states follow a Markov process.

Idiosyncratic risk - income risk

Households also face labor income risk while employed. The logarithm of individual labor income follows an autoregressive process of order one (AR(1)), given by:

$$log(y_{it}) = \mu + \rho log(y_{it-1}) + \xi_{it}$$

$$with \quad \xi_{it} \sim \mathcal{N}(0, \sigma_{\xi}^2)$$

where μ, ρ and σ_{ξ}^2 are the intercept, persistence and variance of the household's income process.

When unemployed, households receive an unemployment benefit ($y_{it} = ub$) with probability p_{ub}. With probability $1 - p_{ub}$, they instead receive a smaller subsistence allowance ($y_{it} = sub$). This reflects the partial coverage of unemployment benefits observed in Italy over the period studied.

[9] The choice of a Cobb-Douglas utility function is motivated by the evidence presented in Piazzesi and Schneider (2007) and Ogaki and Reinhart (1998).

Aggregate risk - employment transitions

In addition to idiosyncratic income and employment risk, households are subject to aggregate risk. The economy alternates between two aggregate states: a good state (boom) and a bad state (recession). Each aggregate state is characterized by a distinct transition matrix governing movements between labor market states for households (permanent employment, fixed-term employment, and unemployment), as well as by different levels of unemployment benefits.

Aggregate risk - wealth shocks

Asset markets are incomplete, and households self-insure against employment and income risk by saving in a composite asset a_t. The composite asset consists of three components: securities, valued at price q_t^s; real estate, valued at price q_t^h; and a risk-free asset (such as deposits and valuables), valued at a fixed price of 1. Households face wealth shocks through fluctuations in the prices of securities and real estate.

The price of securities is subject to aggregate fluctuations, modeled as follows:

$$log(q_t^s) = \rho log(q_{t-1}^s) + \xi_t^s$$
$$with \quad \xi_t^s \sim \mathcal{N}(0, \sigma_{\xi^s}^2)$$

where ρ and $\sigma_{\xi^s}^2$ are persistence and variance parameters.

Similarly, the price of real estate is subject to aggregate fluctuations, modeled as follows:

$$log(q_t^h) = \rho log(q_{t-1}^h) + \xi_t^h$$
$$with \quad \xi_t^h \sim \mathcal{N}(0, \sigma_{\xi^h}^2)$$

where ρ and $\sigma_{\xi^h}^2$ are persistence and variance parameters. I assume that the price processes of securities and real estate are independent.

Finally, the composite asset is bought and sold each period at price q_{jt}

$$q_{jt} = \omega_j^s * q_t^s + \omega_j^h * q_t^h + (1 - \omega_j^s - \omega_j^h) * 1$$

where ω_j^s and ω_j^h represent the weights of securities and real estate in the household's portfolio, respectively. I assume that these weights are exogenous and depend on the household's employment status, denoted by j. Additionally, borrowing is not allowed in this economy.

Durable goods

Households may also self-insure by accumulating durable goods, which are priced at p. The stock of durables, D_t, cannot be negative and depreciates at a rate δ. Moreover, durable purchases are subject to a friction: when households adjust their stock of durables, they incur a non-convex adjustment cost, τ, which is proportional to the stock of durable goods held prior to the adjustment. This adjustment cost follows the specification in Grossman and Laroque (1990), and it reproduces the lumpy patterns of durable consumption documented in the microdata. It can be interpreted as the average resale loss incurred when selling the durable goods on the second-hand market.

Budget constraints

The budget constraint for a household that decides not to adjust its stock of durables is:

$$a_{it}q_{jt} + c_{it} \leq (1+r)q_{jt}a_{it-1} + y_{it}$$

where r is the interest rate on the composite asset.

Conversely, the budget constraint for a household that decides to adjust its stock of durables is:

$$a_{it}q_{jt} + c_{it} + pD_{it} \leq (1+r)q_{jt}a_{it-1} + y_{it} + (1-\tau)(1-\delta)pD_{it-1}$$

The recursive formulation of the problem can be found in Appendix B.

4 Parameterization and Policy Functions

4.1 Parameterization

I parameterize the model to be consistent with key features of the Italian economy before the Great Recession (2002–2006). One period in the model represents a quarter. I use a mixed parameterization strategy: a subset of parameters is fixed using standard values from the literature, while another set is calibrated to match moments from the Italian economy outside the model. The remaining parameters are jointly calibrated using the method of simulated moments within the model. The parameter values are summarized in Table 3, and the targeted moments are shown in Table 4. Details on the numerical implementation can be found in Appendix D.

Calibration of the employment risk

I use the biannual flows of workers between the three labor market states — employed with a permanent contract, employed with a fixed-term contract, and unemployed — from the SHIW data. Using the method of simulated moments,[10] I recover quarterly transitions between contract types and unemployment. I carry out this exercise separately for boom and recession periods.[11] The quarterly transition matrices obtained using this procedure are reported below. Rows represent the current employment state, and columns represent the employment state in the next quarter. 'p' stands for employed with a permanent contract, 'f.t.' for employed with a fixed-term contract, and 'u' for unemployed. For instance, the first row of the left transition matrix indicates that, during a boom, a worker employed with a permanent contract stays in their job with a 98.8% probability, is downgraded to a fixed-term or temporary contract with a 0.08% probability, and loses their job with a probability just under half a percent. These transition matrices are consistent with Lagrosa (2024), who uses Italian administrative data between 1985 and 2019 and reports that approximately 17% of fixed-term workers change their labor status each quarter, compared to less than 2% for workers on permanent contracts.

$$\mathbf{P}_{\mathbf{boom}} = \begin{array}{c c} & \begin{array}{ccc} p & f.t & u \end{array} \\ \begin{array}{c} p \\ f.t \\ u \end{array} & \begin{pmatrix} 0.988 & 0.008 & 0.004 \\ 0.104 & 0.858 & 0.038 \\ 0.029 & 0.042 & 0.929 \end{pmatrix} \end{array} \quad , \quad \mathbf{P}_{\mathbf{recession}} = \begin{array}{c c} & \begin{array}{ccc} p & f.t & u \end{array} \\ \begin{array}{c} p \\ f.t \\ u \end{array} & \begin{pmatrix} 0.984 & 0.010 & 0.006 \\ 0.063 & 0.894 & 0.042 \\ 0.019 & 0.030 & 0.951 \end{pmatrix} \end{array}$$

The first takeaway from these estimates is that permanent contract holders face a significantly lower risk of losing their job than fixed-term contract workers. The unemployment risk at a quarterly time horizon is around 4% for fixed-term contract holders, compared to just half a percent for permanent contract holders. Looking at the flows of workers during booms and recessions, I find that both types of contract experience an increase in the risk of job loss during recessions. However, this is not

[10] I assume that transitions between employment states follow a Markov process. I use a diagonal matrix of the inverses of the moments' relative variances as the weighting matrix. The moments' variances are obtained using a thousand-repetition bootstrapping procedure.Appendix C.1 presents the biannual employment transitions from both the data and the model.

[11] Data on contract types is only available in the SHIW from 2000 onwards. Consequently, I use the SHIW data before 2008 to calibrate the model's quarterly transitions during a boom period, and the SHIW data from 2008 to 2014 to recover the model's transitions during the recession.

the only change for fixed-term workers. Their probability of being upgraded to a permanent contract falls significantly in a downturn (from 10.5% to 6%).

Calibration of the income processes

Income processes are calibrated using SHIW data from 2000 to 2006. In period t, the logarithm of household i 's income $log(y_{it})$ is given by:

$$log(y_{it}) = Z_{it}'\beta + \tilde{y_{it}}$$

$$\tilde{y_{it}} = P_{it} + \epsilon_{it}$$

$$P_{it} = \tilde{\mu} + \tilde{\rho} P_{it-1} + u_{it}$$

$$\epsilon_{it} \sim i.i.d, \quad u_{it} \sim \mathcal{N}(0,\, \sigma_u^2)$$

where Z_{it} is a set of households' observable characteristics, including type of contract and other demographic variables. The income residual $\tilde{y_{it}}$ has a persistent component P_{it}, which follows an autoregressive process of order one, along with some i.i.d. measurement error ϵ_{it}. Income residuals are obtained by performing a standard OLS regression of the logarithm of individuals' labor income on year dummies, type of contract, gender, age, age squared, education, region, and city size. I then use variance-covariance identifying restrictions to recover the persistent component's intercept, persistence, and variance parameters. After discarding measurement error, the income process exhibits quarterly persistence just above 0.98 and a variance of 0.0034. These estimates align with the boom quarterly values reported by Storesletten et al. (2004). Finally, type-of-contract-specific average fitted values from the regression are added to the residuals to obtain the final income grids for permanent and fixed-term workers.[12]

Unemployment benefit levels are set to replicate the mean unemployment transfers of households receiving unemployment benefits. For the boom period, I use SHIW data from 2002 to 2006, and for the recession, I consider observations from 2008 to 2014. The probability of receiving unemployment benefits is chosen to match the SHIW coverage rate for unemployment benefits during the period, which is roughly 12%.

[12] Additional details are provided in Appendix C.2

Calibration of the wealth shocks

Securities price fluctuations are calibrated using the log of the S&P500 adjusted closing prices, deflated by the CPI, from January 1985 to December 2007.[13] A similar procedure is applied to real estate price fluctuations, using the log of the real residential property price index for Italy from January 1980 to December 2007. The weights of securities and real estate in households' portfolios come from SHIW data between 2002 and 2006. For permanent households, the average portfolio consists of 5% securities[14], 59% real estate, and 36% deposits and other real assets.[15] Fixed-term households' portfolios consist of only 1% securities, 46% real estate, and 52% deposits and other real assets. Unemployed households' portfolio weights are set to be the same as those of fixed-term households.

Other parameters fixed outside the model

I set the coefficient of relative risk aversion to 2. The interest rate on the composite asset is set to 0.01, which corresponds to an annual interest rate of approximately 4%. The price of durable goods is normalized to 1. In line with Harmenberg and Oberg (2021) and Attanasio et al. (2022), I set the net depreciation rate of cars to 10.56% per year.[16] The subsistence allowance for households that do not receive unemployment benefits is set to €100 per month. Finally, I set the transitions between booms and recessions to match the average length of recessions (7.5 quarters) and the share of total time spent in recessions (43%) in Italy between 1960 and 2016.[17] All calibrated parameters are summarized in Table 3.

Parameters jointly calibrated inside the model

I jointly calibrate the remaining parameters inside the model using the method of simulated moments on SHIW data from 2002 to 2006.[18] The rich wealth distribu-

[13] After removing a linear trend, I use a standard OLS regression to estimate the AR(1) persistence parameter and the variance of the residuals.

[14] (including government bonds)

[15] Households do not hold any business assets, as self-employed individuals are excluded from the sample.

[16] Harmenberg and Oberg (2021) estimate an annual depreciation rate for durables of 9.2%, while Attanasio et al. (2022) estimate a rate of 11.9% (in the absence of maintenance). I take the average between the two.

[17] Following the method in Krueger, Mitman, and Perri (2016). I use the OECD-based recession indicators for Italy, computed by the Fed of St. Louis.

[18] As with the income process, I use a bootstrapping procedure with a thousand repetitions to obtain the moments' variances. I then take the diagonal matrix of the inverses of the moments'

Table 3: Parameters

Parameter	Value	Description	Target
Households			
β	0.979	Discount factor	See Table 4
σ	2.00	Relative risk aversion	Standard value
r	0.01	Interest rate	Annual interest rate of 4%
α	0.936	Weight of non-durable consumption in utility	See Table 4
τ	0.065	Durables adjustment cost	See Table 4
δ	0.026	Depreciation rate	Harmenberg and Oberg (2021), Attanasio et al. (2022)
p	1.00	Durables price	Normalisation
ub_{boom}	0.32	Unemployment benefit in boom	Mean unemployment benefit 2002-2006
$ub_{recession}$	0.25	Unemployment benefit in recession	Mean unemployment benefit 2008-2014
sub	0.05	Subsistence allowance	€100 per month
p_{ub}	0.12	Probability to get unemployment benefit	Unemployment benefit coverage 2002-2014
Aggregate state			
ρ_{bb}	0.90	Boom to boom transition	Time spent in recession
ρ_{rr}	0.87	Recession to recession transition	Average length of recession

Notes: All values are reported at the quarterly frequency of the model.

tion in the model and the presence of non-convex adjustment costs introduce strong path-dependence. Consequently, extra care must be taken when setting the initial condition in the simulations. Starting from the stationary distribution for durables and liquid wealth, I simulate the model for the 1980–2001 period, feeding in the path of realized aggregate shocks for Italy (OECD-based recession indicators). I then compute the model's moments for the expansion period between 2002 and 2006. This procedure ensures that the moments from the model are comparable with their data counterparts. The jointly calibrated parameters are the households' discount factor (β), the non-convex adjustment cost for cars (τ), and the weight of non-durables in the utility function (α). The set of targets and the associated estimates are displayed in Tables 3 and 4. The calibration results align well with estimates from the literature.

Table 4: Targeted Moments

Target	Model	Data	Source
Share of households buying a car in a year	0.16	0.16	SHIW 2002-2006
Mean car expenses normalized by income	0.33	0.33	SHIW 2002-2006
Median wealth normalized by income	3.55	3.55	SHIW 2002-2006

Notes: Income refers to yearly income from labor and transfers. The mean of car expenses is conditional on buying a car.

relative variances as a weighting matrix.

4.2 Model Fit: Non-targeted Moments

This subsection presents how the model fits important moments that were not explicitly targeted during the calibration. Table 5 displays these cross-sectional moments for both the model and the data. The data corresponds to the years between 2002 and 2006, and the model simulates the same time period, with the procedure for setting the initial condition explained in the previous subsection.

As shown on the top half of Table 5, the model is particularly successful in reproducing non targeted moments by type of contract. This is key as the aim of the paper is to analyze and compare permanent and fixed-term households' consumption patterns over the business cycle. The model reproduces the share of permanent and fixed-term households who adjust their car, as well as ratios of car purchases, income and consumption by type of contract. The detailed calibration of contract specific income and employment risk and of cars adjustment cost explains why these non-targeted income and consumption moments are well aligned with the data. As is common in this type of models, I do not capture the high degree of wealth concentration among the very rich. Consequently the model underestimates permanent households' wealth relative to that of fixed-term households.

The bottom part of Table 5 displays moments for the entire population. The model matches each of these non-targeted moments closely.

4.3 Policy Functions

Figure 1 plots the choices of durable stock as a function of the initial durable stock for permanent (in blue) and fixed-term households (in yellow). These policy functions illustrate that households update their durable stock following a trigger-target (S,s) rule. Since durable investments are subject to non-convex adjustment costs, agents seek to limit the frequency of such adjustments. Households set a minimum stock of durables below which they do not want to fall, and a maximum stock above which their stock is inefficiently high. As long as their durable stock remains between these two bounds, households will make no adjustments and simply allow their durables to depreciate. Once the durable stock falls below the lower trigger point, households become willing to pay the adjustment cost to bring their stock up to the target value. Figure 1 shows the adjustment trigger and target points for buying and selling durables. Between these points, households remain in the inaction region.

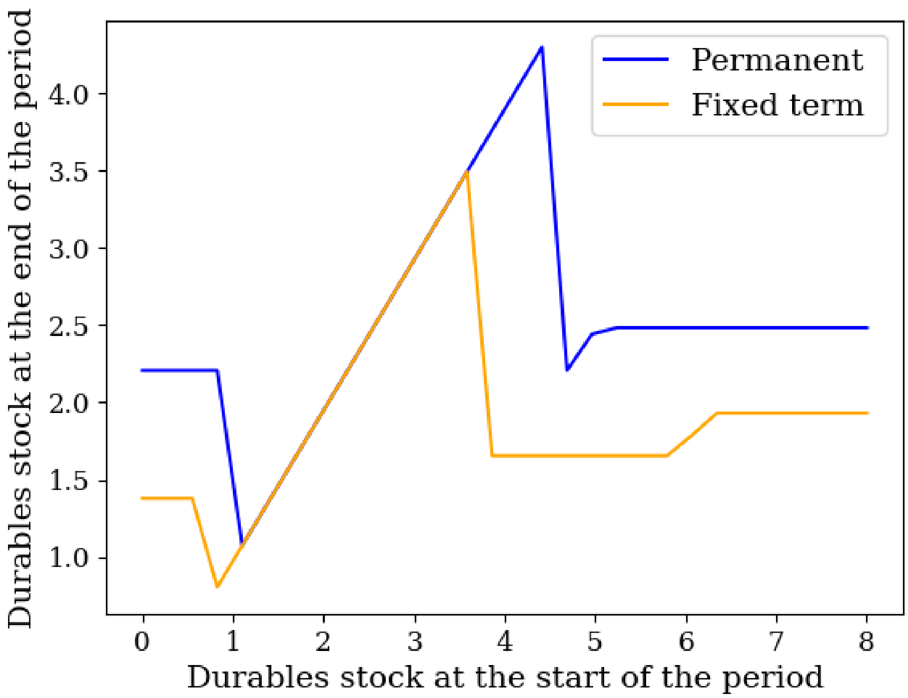
The figure also highlights the differences between permanent and fixed-term households. Permanent households' optimal durable stock is higher than that of fixed-term

Table 5: Non-targeted Moments

Moment	Model	Data
By type of contract		
Share of car adjusters - perm.	0.17	0.16
Share of car adjusters - f.t.	0.13	0.13
Ratio perm./f.t. - car purchases	1.27	1.26
Ratio perm./f.t. - income	1.86	1.86
Ratio perm./f.t. - cons.	1.40	1.39
Ratio perm./f.t. - wealth	1.31	2.00
Overall population		
Mean ratio car purchases / cons.	0.32	0.41
Mean ratio car purchases / stock cars	0.74	0.69
Mean ratio stock cars / cons.	0.42	0.47
Mean share of cars in total assets	0.11	0.07
Share of car downgraders	0.004	0.006

Notes: Cons. stands for non-durable consumption. Ratio perm./f.t. refers to the mean of the variable for permanent households over the mean of the variable for fixed-term households. Income refers to income from labor and transfers. Car purchases are conditional on buying a car. Total assets refer to wealth + stock of cars.

Figure 1: Policy Function: Stock of Durables



Notes: Choices of durable stock as a function of the initial durable stock. The other states are held fixed at the mean cash-in-hand, mean income, boom aggregate state for employment risk, and median prices for securities and real estate.

households. Furthermore, fixed-term households will wait longer before purchasing new durables and will sell their existing stock more quickly than their permanent counterparts. As a result, the inaction region for fixed-term households is shifted downward.

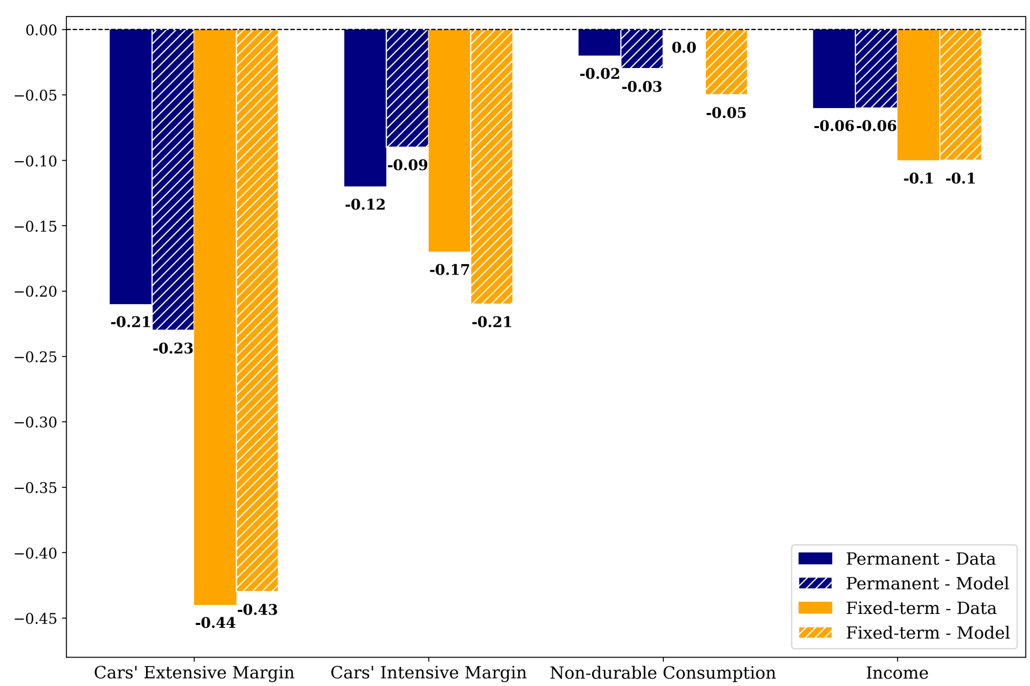
5 The Recession Experiment

5.1 Baseline

I simulate the model's response to a shock calibrated to reproduce the Great Recession and analyze the consumption patterns of durable and non-durable goods for both permanent and fixed-term households. As in the calibration procedure, I carefully set the simulations' initial conditions, given that durable purchases are pathdependent. The simulations begin with real-world booms and recessions between 1980 and 2001, using Italy's OECD-based recession indicators. Next, I simulate a boom for the 2002–2007 period, where securities prices are held at the median and real estate prices are high (to match the surge in real estate values before the Great Recession). The simulation then moves to the Great Recession (2008–2014), incorporating recession-specific employment risk, low securities and real estate prices, as well as an additional MIT shock to labor income, calibrated to reflect the income drop experienced by both fixed-term and permanent households during this time.

The aim of this exercise is to assess whether the model can replicate the empirical patterns discussed in the motivation section of the paper. Therefore, the data sample selection used in the empirical analysis should be mirrored in the model simulations. In the data, households report their employment status for most of the year. As a result, households classified as fixed-term (permanent) could have been employed with a permanent (fixed-term) contract or unemployed for a brief period during the year. The same approach applies to the model simulations.

Figure 2: Recession Experiment



Notes: Changes in car purchases, non-durable consumption, and income during the Great Recession for permanent and fixed-term households (model vs. data).

The results of the recession experiment are shown in Figure 2, where the bars represent changes in the extensive margin of car purchases, the intensive margin of car purchases, non-durable consumption, and income over the course of the Great Recession for permanent (blue bars) and fixed-term (yellow bars) households. The solid bars correspond to the motivational evidence presented in Table 1, while the dashed bars represent the model simulations. Each household type's drop in labor income is a targeted moment, while the changes in car and non-durable consumption are not. The model successfully replicates the patterns observed during the Great Recession. Over this period, the model's share of households purchasing a car dropped by 23% and 43% for permanent and fixed-term households, respectively, compared to 21% and 44% in the data. Moreover, the intensive margin of car purchases for fixed-term households decreased by 21% in the model, compared to 17% in the data. For permanent contract holders, these figures are 9% and 12%, respectively. Finally, the model closely tracks the change in non-durable consumption for permanent households, while slightly overestimating the contraction for fixed-term households.

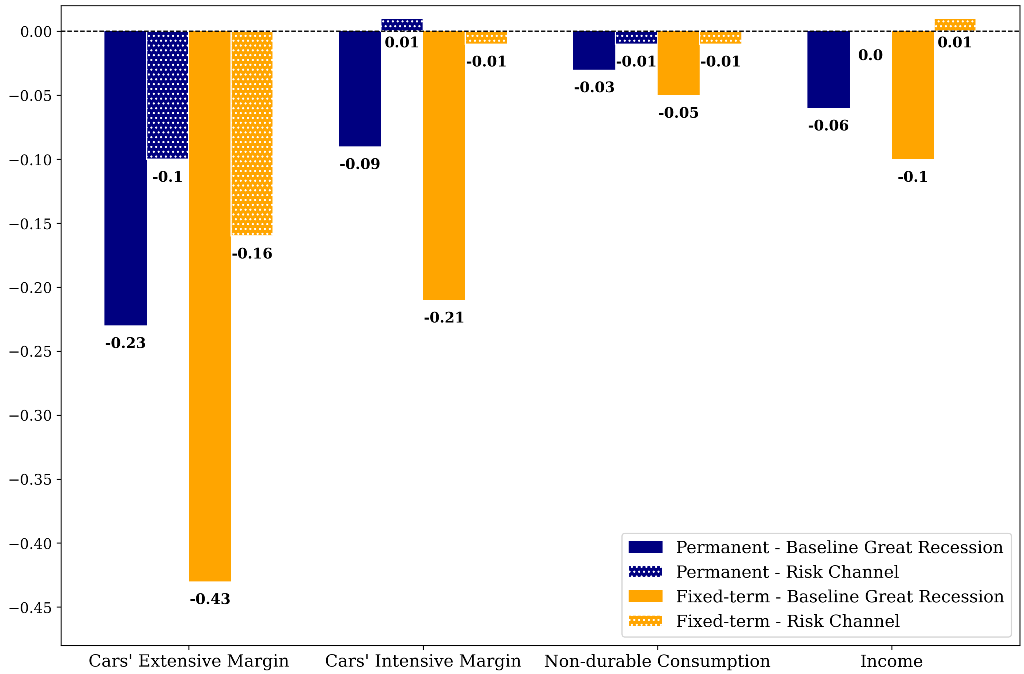
5.2 The Role of Income Risk

The previous subsection established that the model successfully replicates the observed consumption patterns of permanent and fixed-term households during the Great Recession. I now aim to disentangle the impact of changes in household's income risk from that of realized income losses. To do this, I compare the baseline recession experiment to a placebo experiment.[19] In the placebo experiment, households believe that they are in a recession (using their recession policy functions), but the underlying shocks to the economy are expansionary. Specifically, the aggregate state for employment risk is a boom, securities and real estate prices are set to those of the 2002–2006 period, and there is no additional unexpected labor income drop. The placebo experiment highlights the risk channel. The results of this decomposition are shown in Figure 3, where the solid bars correspond to the model's baseline recession experiment, and the dotted bars illustrate the placebo counterfactual (risk channel).

In the placebo scenario, households' income remains unchanged. For permanent households, changes in income risk account for 43% the baseline decline in the extensive margin of car purchases, but none of the decline in the intensive margin. For fixed-term households, the change in income risk accounts for 38% of their baseline decline in car purchases on the extensive margin and 5% on the intensive margin.

[19] In the spirit of Harmenberg and Oberg (2021).

Figure 3: Recession Experiment: Risk Channel



Notes: Changes in car purchases, non-durable consumption, and income (Great Recession vs. placebo). In the placebo experiment, households believe that they are in a recession (using their recession policy functions), but the underlying shocks to the economy are expansionary. The placebo experiment highlights the impact of changes in risk: the risk channel.

Permanent households: downside income risk

Regardless of whether households hold fixed term or permanent contracts, the income risk channel plays a significant role in explaining the decline in the extensive margin of car consumption during the Great Recession. However, the mechanisms behind the risk channel are fundamentally different. To illustrate, Table 6 presents moments of the income distribution during the boom and the recession, broken down by contract type.[20]

First, we consider permanent households. Consistent with Guvenen et al. (2021), the income distribution is negatively skewed. Upon entering the Great Recession, permanent households experience an increased probability of unemployment (from 0.4% to 0.6%) and of transitioning to a fixed-term contract (from 0.8% to 1%). As a result, their income variance rises by nearly a quarter. This leads to an expansion

[20] These moments are computed without the additional Great Recession labor income drop.

Table 6: Moments of Households' Income Distribution

	Mean		**Variance**		**Skewness**	
	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.
Boom	1.035	0.543	0.011	0.049	-4.161	2.257
Recession	1.032	0.520	0.014	0.035	-4.465	2.364
Change	0.0	-0.04	0.25	-0.29	0.07	0.05

Notes: Perm. stands for permanent contract and F.t. for fixed-term or temporary contract. Boom is 2002-2006, Recession is 2008-2014. Income is income from labor and transfers.

in the lower tail of the income distribution, as permanent households face a higher risk of large income declines. This shift in income risk interacts with car expenses because the non-convex adjustment cost makes car purchases partially irreversible. In other words, purchasing a car becomes a commitment that permanent households are less willing to make in an environment of heightened downside income risk. In such conditions, it is optimal for them to delay durable purchases until uncertainty subsides. Higher income uncertainty encourages permanent households to "Wait-to-See", which in turn widens the inactivity region of their policy function. The widening of the inactivity region is associated with a downward shift in the trigger adjustment point (s). Since the risk channel does not lead to changes on the intensive margin of car purchases, we cannot infer that the optimal durable stock (S) has changed. To sum up, the rise in downside income risk drives permanent households' car purchases response during the Great Recession. The associated strategy is "Wait-to-See".

Fixed-term households: upside income risk

Table 6 shows that the income distribution of fixed-term households is positively skewed. This is because, for them, a source of large positive income changes is the possibility of being upgraded to a permanent contract. In other words, fixed-term households have a lot of room to move up and upside income risk is an important feature of their income process. This is consistent with Almuzara et al. (2025) who find that the conditional skewness of income shifts from positive to negative along the distribution of persistent income.

Upon entering the Great Recession, the income process for fixed-term households shifts in two ways. First, similar to their permanent counterparts, fixed-term households' unemployment hazard increases (from 3.8% to 4.2%). Second, their probability of being upgraded to a permanent contract decreases significantly (from 10.4% to 6.3%), compressing the upper tail of the income distribution. Overall, the second channel dominates, as fixed-term households experience a 30% drop in income variance during the Great Recession.

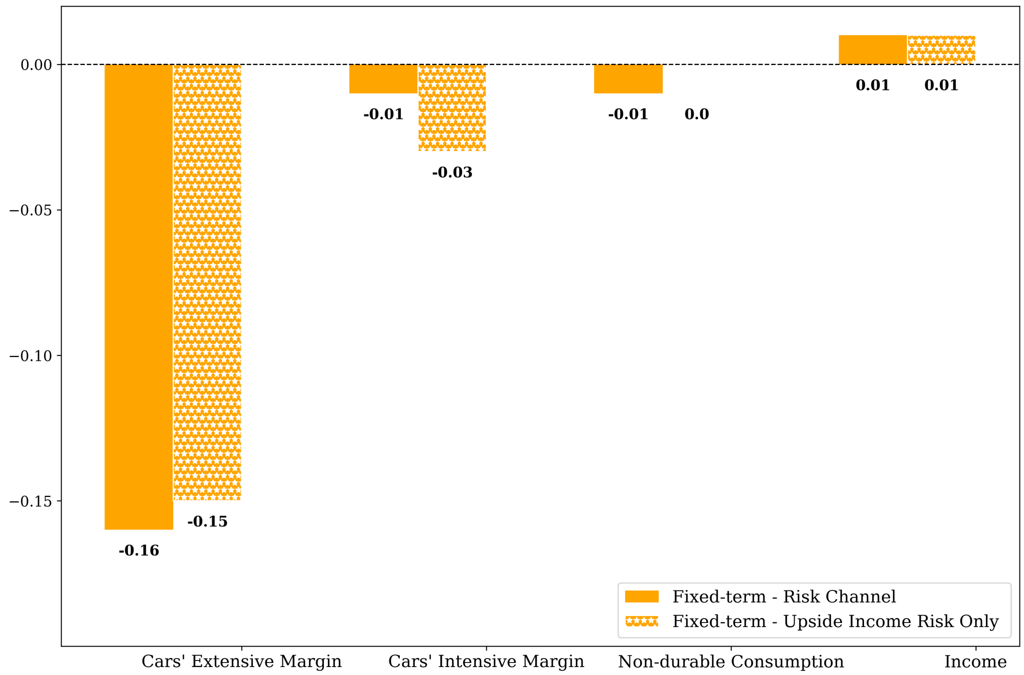
In terms of durable consumption, the standard "Wait-to-See" story is a poor fit for fixed-term households' behavior, as they do not need to wait for the uncertainty to subside—it already has. To understand the mechanism at play, it is important to note that, alongside the decline in the extensive margin of car purchases observed in the placebo experiment, the intensive margin of car consumption also contracts. This change is not trivial, as a mechanical force would predict the opposite outcome. As households wait longer to upgrade their vehicles, the value of their cars depreciates further. Consequently, the gap between the car before adjustment and the optimal car becomes larger during the recession, suggesting that the average size of net purchases should be larger. The fact that not only did the average car purchase fail to rise, but actually decreased during this period, highlights that fixed-term households adjusted their optimal stock of durables (S) downward. As upgrading to a permanent contract becomes less likely, fixed-term households experience greater persistence in their high-risk, low-income employment status. As a result, they seek to rebalance their portfolio away from durable goods, since partial irreversibility makes them a poor hedge against labor income fluctuations. This result is in line with Attanasio et al. (2022), who find that a flattening of the life-cycle profile triggers a contraction in consumption on the intensive margin of car purchases.

To rebalance their portfolio, fixed-term households seek to downsize their cars, but the non-convex adjustment cost make selling durables costly. Furthermore, because the cost is non-convex, it is particularly discouraging for low-income, low-wealth households, who are overrepresented among those with fixed-term contracts. This is the case because the fixed cost of adjusting durables represents a larger fraction of their resources. As depreciation naturally reduces the value of cars over time, fixed-term households rely on this mechanism to downsize and save on transaction costs. In essence, they 'Wait-to-Rebalance.'

The literature has focused on how a rise in income variance can lead households to become stuck within the inaction band of durable consumption. Here, I show that if the change in risk stems from a collapse in upside income risk, a decrease in income variance can also widen households' durable consumption inaction band.

To further test this mechanism, I run a placebo counterfactual in which I neutralize the rise in unemployment risk for fixed-term households. In this exercise, the only change in the income process comes from the drop in the probability of being upgraded to a permanent contract, i.e., the reduction in upside income risk. Fig-

Figure 4: Recession Experiment: Upside Risk Only (Fixed-term)



Notes: Changes in car purchases, non-durable consumption, and income for fixed-term households. I plot the placebo experiment (i.e., the risk channel) alongside a placebo experiment in which the only change in risk comes from a collapse in upside income risk (i.e., upside risk only). In this case, downside income risk remains the same as in boom.

ure 4 plots this upside income risk only counterfactual (in starred bars) alongside the baseline placebo experiment from Figure 3 (in solid bars). The baseline drop in car consumption persists across both margins when the decline in upside income risk is the only force at play.[21] To sum up, the drop in upside income risk drives fixed-term households' car consumption response during the recession. The associated strategy is "Wait-to-Rebalance".

[21] If we run the opposite experiment, where we only keep the rise in unemployment risk (downside income risk), the contraction in car consumption on the intensive margin disappears. Some drop in the probability of purchasing a car subsides and can be attributed to a "Wait-to-See" mechanism, similar to what we observe in the permanent contract case.

6 "Wait-to-See" *versus* "Wait-to-Rebalance"

We have just established that both permanent and fixed-term households reduce their durable purchases during the recession; however, the mechanisms driving these adjustments are significantly different. In this subsection, I investigate how disentangling these mechanisms and distinguishing between contract types matters when evaluating the persistence of consumption contraction, as well as the implications for public policy.

6.1 Persistence of Consumption Contraction

The driver of the drop in durable goods purchases — specifically between upside versus downside income risk - has a significant impact on the duration of the contraction. Figure 5 plots changes in the extensive margin of car purchases for permanent households (in blue) and fixed-term households (in yellow). The baseline period is 2007,[22] with the Great Recession beginning in 2008[23] and continuing throughout the plotted period.

For permanent households, the drop in car purchases is relatively short-lived. As the recession progresses, depreciation lowers the value of households' existing cars. Eventually, this value declines enough that permanent households are forced to adjust, despite the ongoing recession. The "Wait-to-See" strategy triggers a sharp but short-lived decline in car purchases. In contrast, fixed-term households' "Wait-to-Rebalance" is driven by the slow process of depreciation and results in a very persistent decline throughout the recession.

6.2 Policy Experiment: Car Purchase Subsidy

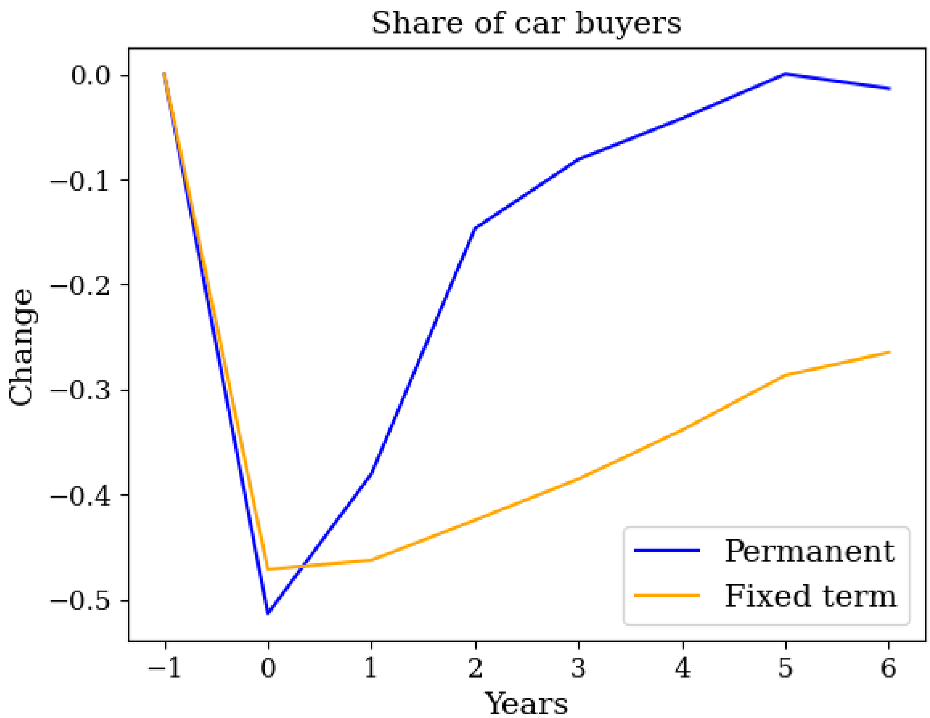
Does taking into account the dual structure of the labor market matter for designing fiscal stimulus? In 2009, the Italian government allocated 2 billion euros in subsidies to support the automobile sector. I use the model as a laboratory to investigate the impact of a payment corresponding to 5% of car expenses during that year. For the average car purchase, this translates to approximately €500. In the simulations, households do not anticipate the implementation of the subsidy. Besides, while they are aware of the program in 2009, they do not expect it to last for only one year.

The left panel of Figure 6 shows the share of households that bought a car in a given

[22] t = -1 on the plot.

[23] t = 0 on the plot

Figure 5: Persistence of "Wait-to-See" *versus* "Wait-to-Rebalance"

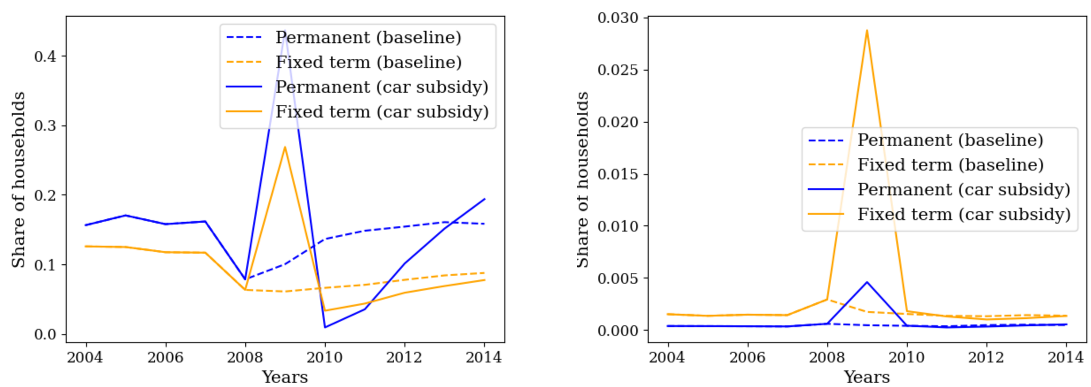


Notes: Changes in the extensive margin of car purchases. The baseline period is 2007 ($t = -1$), with the Great Recession beginning in 2008 ($t = 0$) and continuing throughout the plotted period.

year, for both the permanent group (in blue) and the fixed-term group (in yellow). The solid lines represent the simulations with the 2009 car subsidy, while the dashed lines represent the baseline experiment during the Great Recession. In 2009, the subsidy significantly increases the share of buyers (44% for permanent households and 27% for fixed-term households). However, once the policy ends, there is a sharp reversal, with the share of buyers dropping substantially in the following years. The car subsidy did not lead to net new purchases, but rather encouraged households to bring forward their car investments. As a result, car consumption was depressed in the post-program period. This finding is consistent with the empirical analyses of Mian and Sufi (2012), Hoekstra, et al. (2017), and Green et al. (2020), who study the cash-for-clunkers policy in the US and find that its stimulative effect on car consumption primarily stems from changes in the timing of purchases, rather than inducing new purchases.

Additionally, the driving forces behind consumption contraction must be carefully considered when designing such fiscal policies. The impact of the subsidy depends on households' type of contract. In 2009, the spike in car purchases among fixed-term households is milder compared to their permanent counterparts. Fixed-term households, who implement a "Wait-to-Rebalance" strategy, wish to reduce their car holdings, meaning the subsidy has limited ability to encourage upgrades. What is more, the subsidy can even be viewed as a downgrading subsidy. The right panel of Figure 6 shows the share of households who downgraded their car in a given year (i.e., sold their car to purchase a cheaper one). In 2009, a portion of fixed-term households took advantage of the subsidy to downgrade their car holdings. The subsidy is used by fixed-term households to cover transaction costs. In this sense, it is related to Gavazza and Lanteri (2021), where the adjustment cost is endogenous and depends on the equilibrium price of used cars.

Figure 6: Car Purchase Subsidy



(a) Share of Buyers (b) Share of Sellers

Notes: Model-simulated share of households buying a car (panel a) and selling a car (panel b) between 2004 and 2014. The dashed lines correspond to the baseline Great Recession experiment, while the solid lines represent a 5% car purchase subsidy in 2009.

7 Labor Market Policy: Contract Harmonization

The final section of this paper uses the model to explore how labor market policy interacts with aggregate stabilization. I examine the impact of the Italian-style two-tier segmentation of the labor market, which involves distinct types of contracts, each associated with different levels of employment protection legislation (EPL). EPL refers to the strictness of hiring and firing regulations. Dual-EPL systems introduce flexibility at the margin, creating an EPL gap between different types of workers.[24]

A vast body of literature has examined the effects of reforms aimed at facilitating the use of fixed-term contracts while leaving regulations for permanent contracts unchanged. These reforms widen the EPL gap.[25] Facilitating the creation of temporary jobs promotes job creation but also increases job destruction, leading to a potentially ambiguous effect on unemployment. Boeri and Garibaldi (2007) identify what they call a 'honeymoon effect,' with transitory employment gains that are eventually offset by the decline of insider workers. Several studies have found that EPL gap-widening reforms lead to decreased productivity and fail to increase employment (Blanchard and Landier, 2002; Bentolila and Saint-Paul, 1992; Cahuc et al., 2016; Cappellari et al., 2012; Daruich et al., 2023; Dolado et al., 2021; Hijzen et al., 2017). The mechanism at play is as follows: the larger the EPL gap, the higher the rate of hiring on fixed-term contracts and the lower the rate of conversion into permanent contracts. This leads to a large-scale replacement of permanent jobs with temporary ones. Consistent with this, our empirical estimates using SHIW data indicate that the probability of fixed-term contracts being upgraded to permanent ones significantly decreased after 2008.

Additionally, the literature documents large increases in employment volatility following EPL-widening reforms. For instance, Costain et al. (2010) find that unemployment fluctuates 33% more under EPL duality than in a unified economy with the same average unemployment rate. Similarly, Bentolila et al. (2012) observe that France and Spain had similar unemployment rates of around 8% just before the Great Recession, but these rates subsequently increased to 10% in France and 23% in Spain. The EPL gap is wider in Spain, and the authors find that Spain could have avoided about 45% of its unemployment surge had it adopted the French employment protection legislation.[26] For these reasons, several studies advocate for versions of unified open-ended contracts (Andrés et al., 2009; Blanchard and Tirole, 2008; Boeri

[24]Permanent contracts can only be terminated for disciplinary or economic reasons, with severance pay due only in the latter case. Additionally, workers can appeal their dismissal in court. In contrast, fixed-term contracts often do not include severance pay upon expiration, and workers cannot appeal the termination in court.

[25]In the Italian context, an example of such a reform is the September 2001 reform, which replaced the strict set of conditions allowing temporary contracts with a general provision permitting employers to use temporary employment for any 'technical, production, organizational, or substitution' reasons. See Boeri (2011) for a survey of institutional reforms and labor market dualism in OECD countries.

[26]See Cahuc and Postel-Vinay (2002) and Sala and Toledo (2012) for similar arguments.

and Garibaldi, 2008; Cahuc, 2012; Lepage-Saucier et al., 2013), and policymakers have taken steps in this direction.[27]

In light of these evidence, I now use the model to compare the depth of the consumption contraction during the Great Recession between the baseline dual-EPL Italian economy and the limit case of an economy with a single EPL. In the single-EPL counterfactual, I assume there is only one type of contract in the labor market and use the SHIW data to calibrate workers' income risk, as well as state-dependent transitions in and out of employment.[28] This implies that the average level of risk in the economy remains unchanged but is uniformly shared by all workers, resulting in a single contract that bears the average risk. The results are displayed in Table 7, where the baseline dual-EPL consumption drop during the Great Recession is normalized to -1. I report changes in the extensive and the intensive margins of car purchases over the Great Recession for employed workers.[29]

Table 7: Durable Consumption Contractions: Alternative EPL

	Extensive margin	Intensive margin
Dual EPL (baseline)	-1	-1
Single EPL	-1.1	-0.87
Single EPL: -30% unemployment rise	-0.93	-0.95
Single EPL: -45% unemployment rise	-0.87	-0.96

Notes: Changes in the extensive margin and the intensive margin of car consumption. I normalize the consumption contraction in the baseline two-contract (dual EPL) Great Recession simulation to -1. I also report the consumption drop for the Great Recession simulations under a harmonized single labor contract (single EPL). In the single EPL simulations, I consider three cases: the unemployment volatility remains unchanged, decreases by 30% following Costain et al. (2010), and decreases by 45% following Bentolila et al. (2012).

By comparing the first two rows, we see that the effects of moving to a single-EPL labor market would have been mixed during the Great Recession. In this case, the contraction in car consumption is about 13% smaller on the intensive margin but about 10% larger on the extensive margin, relative to the two-contract baseline. This ambiguous effect of closing the EPL gap between fixed-term and permanent workers, and creating a unified contract, is driven by two opposing forces. On the one hand,

[27] In Italy, for instance, the 2015 Jobs Act introduced a new type of permanent contract — the fast-track contract — which offers gradual protection for new employees, with an increasing tenure profile. Boeri and Garibaldi (2018), as well as Sestito and Viviano (2018), find a positive employment effect from the introduction of the fast-track contract.

[28] The calibration procedure is similar to that for the case with two types of contracts. Estimates for the single contract case can be found in the Appendix C.3

[29] The two contract types are pooled together in the dual EPL baseline.

temporary workers drastically reduce their consumption of durable goods during the recession, following the collapse of their upside income risk. Moreover, their fall in demand is very persistent, lasting throughout the recession. Eliminating this type of contract should therefore significantly dampen the contraction in durable consumption during the recession. On the other hand, in the dual-EPL economy, most of the risk is borne by a small segment of the workforce: the fixed-term contract holders. These workers are typically at the bottom of the income and wealth distributions, and therefore account for only a small fraction of aggregate consumption. Conversely, permanent workers, who are responsible for the lion's share of aggregate purchases, are somewhat shielded from the rise in employment risk and maintain their consumption.

Still, the previous comparison assumes that changes in EPL have no impact on employment. Based on the evidence mentioned above, I explore two additional counterfactual experiments in which eliminating the dual-EPL system reduces unemployment volatility. The third row corresponds to a single-EPL counterfactual in which, following Costain et al. (2010), I assume that the rise in unemployment during the Great Recession is 30% milder than in the dual-EPL economy. The fourth row represents the experiment where, drawing on Bentolila et al. (2012), I assume that the unemployment rise in the one-contract economy during the Great Recession is 45% lower than in the baseline dual-EPL economy. Accounting for the potential impact of EPL harmonization on unemployment volatility significantly increases the aggregate stabilizing impact of the reform. In the 45% experiment, the extensive and intensive margins of car purchases shrink by 13% and 4% less, respectively, than in the dual-EPL case. In the 30% experiment, the drop in car consumption is 7% smaller on the extensive margin and 5% smaller on the intensive margin.[30]

8 Conclusion

This paper studies the impact of risk heterogeneity on business cycle volatility through the lens of durable consumption. I analyze risk heterogeneity with a focus on labor contracts and find that, following a negative aggregate shock, workers on fixed-term contracts reduce their durable purchases significantly more than their permanent contract counterparts. Using an incomplete markets model with durable

[30]We note that these reductions in the unemployment surge during the recession are conservative in the sense that, in these scenarios, the regulations around permanent contracts (which account for the majority of the workforce) have been relaxed. As a result, firms would have greater flexibility to thrive.

consumption and a dual labor market, I examine the drivers of consumption changes across contract types. I find that permanent contract workers respond to an increase in their downside income risk, while fixed-term contract holders react mainly to a collapse in their upside income risk. The segment of the income distribution from which the change in risk originates significantly affects both the magnitude and persistence of durable consumption contractions. This insight extends beyond dual-contract labor markets, provided that workers face changes in risk at different ends of the income distribution. This result could also generalize beyond households. Firms, too, engage in lumpy investment, and their propensity to invest over the business cycle may similarly depend on whether they face upside or downside risk. This channel could give rise to interesting amplification mechanisms, which I leave for future research.

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A Data Appendix

A.1 SHIW Data

This appendix provides additional information on the Survey on Household Income and Wealth (SHIW), conducted by the Bank of Italy between 1965 and 2022 (the most recent available wave). I restrict the analysis to the period 2000–2014, as the SHIW did not collect the variables required for my study before 2000, and my focus is on the Great Recession period. Starting in 2000, the SHIW includes detailed information on job-related characteristics, as well as comprehensive data on household income, wealth, non-durable consumption, and durable goods expenditures. The survey also reports data on vehicle purchases and sales.

The SHIW is conducted biennially and includes a panel component, which has expanded over time and accounted for approximately 55% of the sample in 2014. The units of observation are both individuals and households. Each wave comprises around $8,000$ households and $20,000$ individuals. The survey provides sampling weights that make the data representative of Italian households, along with monetary revaluation coefficients computed by Istat.

A.2 Sample Selection

Sample selection: household-level by type of contract. I restrict my analysis to households with at least one employed income earner aged between 20 and 65, excluding households with any self-employed or currently unemployed members. Additionally, I focus on households in which all employed members have the same type of employment contract or households with a single income earner. This restriction ensures a clean classification into either the permanent or fixed-term contract group. Households with mixed contract types or missing information on contract type are excluded from the analysis.

In boom periods, the sample includes approximately 500 households in the fixed-term group and 7,500 in the permanent group. During recessions, the fixed-term and permanent groups include around 1,050 and 9,050 households, respectively. These samples are used for all household-level variables disaggregated by contract type.

However, an exception is made when computing the intensive margin of car purchases for fixed-term households. To ensure a sufficient number of observations, I include any household with at least one member holding a fixed-term contract, including those with two earners under different contract types.

Sample selection: household-level, all employment status. To compute population-wide moments, I restrict the analysis to households with at least one employed or unemployed member aged between 20 and 65 years. Households with any self-employed members are excluded from the sample.

Sample selection: individual-level. To study income and employment state transitions, I include individuals who are currently employed or unemployed, aged between 20 and 65 years, and who have been surveyed in at least two consecutive waves (three consecutive waves when estimating the income risk process conditional on being employed). Observations with missing information on the type of contract or missing income are excluded.

Sample selection: unemployed individuals. To recover the value and coverage of unemployment benefits, I select individuals who are currently unemployed and aged between 20 and 65 years. First-time job seekers are excluded from the sample, as they do not qualify for unemployment benefits.

A.3 Variable Treatment and Definitions

To remove outliers, I winsorize the top and bottom 1% of the distribution for all numerical variables, separately by boom/recession period and by type of contract. For household-level variables, I apply the appropriate household sampling weights. Car and non-durable consumption expenditures are deflated using the Consumer Price Index (CPI) for new cars and the CPI for non-durable consumption, respectively. Other nominal variables are deflated using the Istat revaluation index.

Definitions:

- **Car purchases (or stock)** include all means of transport, such as motorbikes and boats. The data distinguishes between cars and other vehicles only from 2012 onward. For consistency, I treat cars and other vehicles as a single category for the entire period. This should not affect the results, as cars accounted for more than 85% of all vehicles purchased by households between 2012 and 2014. Car purchases (or stock) are normalized by dividing by the OECD scale adult equivalent units.
- **Non-durable consumption** refers to all consumption expenditures excluding durable goods, such as cars, furniture, and appliances. Non-durable consumption is normalized by dividing by OECD scale adult equivalent units.

A.4 Summary Statistics

Table 8 presents household-level summary statistics for the period 2002-2006 (before the Great Recession). The average income and wealth of permanent households is approximately twice that of fixed-term households. Heads of fixed-term households are, on average, younger and less educated than those of permanent households. Additionally, the share of fixed-term households is higher in the poorer regions of Southern Italy.

Table 8: Summary Statistics

	Permanent contract	Fixed-term contract
Mean income (€)	42398	22540
Mean wealth (€)	205756	103075
Mean age of head of hh	44	40
Share living in the North (%)	53	32
Share living in the Center (%)	21	13
Share living in the South (%)	26	55
Share of college graduates (%)	11	6
Share of high school graduates (%)	46	26
Share living in a small city i.e. - 40,000 inhab. (%)	57	60
Mean number of children	1.23	1.33

B Recursive Formulation of the Problem

Let V denote the value function of a household. For notational simplicity, the subscript i is omitted.

$$V(m, D, y, j; \Omega) = max\{V^{keep}(m, D, y, j; \Omega), V^{adj}(x, y, j; \Omega)\}$$

$$s.t \quad x = m + (1-\tau)(1-\delta)pD$$

Where V^{keep} denotes the value function of a household that chooses not to adjust its stock of durables, and V^{adj} the value function of a household that chooses to adjust. m represents cash-on-hand. x is the cash-on-hand available after selling the beginning-of-period stock of durables D. y denotes income, j is the household's employment status, and Ω is the vector of aggregate states, which includes the current aggregate state (boom or recession), the price of securities q^s, and the price of real estate q^h.

The keeper's problem is:

$$V^{keep}(m, D, y, j; \Omega) = \max_{a,c} U(c, D') + \beta E\left[V(m', D', y', j'; \Omega')\right]$$

$$s.t \quad qa = m - c$$

$$D' = (1-\delta)D$$

$$m' = (1+r)q'a + y'$$

$$q = \omega_j^s q^s + \omega_j^h q^h + (1 - \omega_j^s - \omega_j^h)$$
$$q' = \omega_{j'}^s q^{s\prime} + \omega_{j'}^h q^{h\prime} + (1 - \omega_{j'}^s - \omega_{j'}^h)$$
$$y' \sim \Upsilon(y)$$
$$j' \sim \Phi(j)$$
$$\Omega' \sim \Gamma(\Omega)$$

where Υ denotes the conditional distribution of idiosyncratic labor income, Φ is the conditional distribution of employment status, and Γ represents the conditional distribution of the aggregate states.

The adjuster's problem is:

$$V^{adj}(x, y, j; \Omega) = \max_{a,c,D'} U(c, D') + \beta E\left[V(m', D', y', j'; \Omega')\right]$$

$$s.t \quad qa = x - c - pD'$$
$$m' = (1 + r)q'a + y'$$
$$q = \omega_j^s q^s + \omega_j^h q^h + (1 - \omega_j^s - \omega_j^h)$$
$$q' = \omega_{j'}^s q^{s\prime} + \omega_{j'}^h q^{h\prime} + (1 - \omega_{j'}^s - \omega_{j'}^h)$$
$$y' \sim \Upsilon(y)$$
$$j' \sim \Phi(j)$$
$$\Omega' \sim \Gamma(\Omega)$$

Following the nested structure in Druedahl (2021), the adjuster's problem can be formulated as a sequential decision problem. The household first chooses how much durable goods to buy or sell, and then chooses non-durable consumption conditional on that durables choice. I rewrite the adjuster's problem as follows:

$$V^{adj}(x, y, j; \Omega) = \max_{D'} V^{keep}(m, D, y, j; \Omega)$$
$$s.t \quad D' = (1 - \delta)D$$
$$m = x - pD'$$

C Calibration

C.1 Employment State Transitions

Table 9: Targeted Moments of the Employment Transitions

Boom

Target Transitions (2-year time horizon)	Data	Model
Perm. to Perm.	0.93	0.93
Perm. to F.T.	0.04	0.04
Perm to Unem.	0.03	0.03
F.T. to Perm.	0.49	0.49
F.T to F.T	0.36	0.36
F.T. to Unem.	0.15	0.15
Unem. to Perm.	0.23	0.23
Unem. to F.T.	0.15	0.15
Unem. to Unem.	0.62	0.62

Recession

Target Transitions (2-year time horizon)	Data	Model
Perm. to Perm.	0.90	0.90
Perm. to F.T.	0.06	0.06
Perm to Unem.	0.05	0.05
F.T. to Perm.	0.33	0.33
F.T to F.T	0.47	0.47
F.T. to Unem.	0.20	0.20
Unem. to Perm.	0.14	0.14
Unem. to F.T.	0.13	0.13
Unem. to Unem.	0.73	0.73

Notes: Note that the 'Perm. to Perm.' transition in the first column does not imply that the worker remained in permanent employment for two full years. It simply indicates that, at the two survey dates, the worker was employed under a permanent contract. The worker could have experienced other employment states between the two surveys.

C.2 Calibration of the Income Risk

I use the following restrictions to identify the persistent component's intercept, persistence and variance parameters:

$$\frac{Cov(\tilde{y_{it}}, \tilde{y_{it-2}})}{Cov(\tilde{y_{it}}, \tilde{y_{it-1}})} = \tilde{\rho}$$

$$Cov(\tilde{y_{it}}, \tilde{y_{it-1}}) = \tilde{\rho} * \sigma_P^2$$

$$(1 - \tilde{rho}^2) * \sigma_P^2 = \sigma_u^2$$

$$E(\tilde{y_{it}}) = \frac{\tilde{\mu}}{1 - \tilde{\rho}}$$

The normalized income grids for each type of contract are presented below.

Permanent contract:

$$\textbf{Income} = \begin{array}{ccccc} y1 & y2 & y3 & y4 & y5 \\ (0.55 & 0.74 & 1 & 1.34 & 1.81) \end{array}$$

Temporary/fixed-term contract:

$$\textbf{Income} = \begin{array}{ccccc} y1 & y2 & y3 & y4 & y5 \\ (0.27 & 0.36 & 0.48 & 0.65 & 0.87) \end{array}$$

C.3 Income and Employment Risk: One Contract

This appendix presents the calibrated employment risk and income grid used in the the Section 7 experiments. In these experiments, there is only one type of contract in the labor market. I use the SHIW data to calibrate workers' income risk, as well as state-dependent transitions in and out of employment. The state-dependent transition matrices and the income grid for employed workers are shown below.

$$
\mathbf{P}_{\mathbf{boom}} = \begin{array}{c c} & \begin{array}{cc} e & u \end{array} \\ \begin{array}{c} e \\ u \end{array} & \begin{pmatrix} 0.994 & 0.006 \\ 0.068 & 0.932 \end{pmatrix} \end{array} \quad , \quad \mathbf{P}_{\mathbf{recession}} = \begin{array}{c c} & \begin{array}{cc} e & u \end{array} \\ \begin{array}{c} e \\ u \end{array} & \begin{pmatrix} 0.989 & 0.011 \\ 0.046 & 0.954 \end{pmatrix} \end{array}
$$

$$
\mathbf{Income} = \begin{array}{c} \begin{array}{ccccc} y1 & y2 & y3 & y4 & y5 \end{array} \\ \begin{pmatrix} 0.53 & 0.72 & 0.97 & 1.30 & 1.74 \end{pmatrix} \end{array}
$$

D Numerical Implementation

I solve for the model's policy functions using the NEGM+ algorithm developed by Druedahl (2021). This algorithm extends Carroll (2006)'s endogenous grid-point method to economies with non-convexities and leverages the nested structure of the problem. Additionally, an enhanced interpolation method introduces an extra layer of optimization.

I solve for households' policies on a 30-point grid for durable goods stock and 200-point grids for regular cash-in-hand, cash-in-hand after reselling the stock of durables, and liquid assets. Since these three grids represent the same dimension (cash-in-hand, cash-in-hand after reselling durables, and liquid assets), special care is required in configuring them relative to each other. Specifically, the upper bound of the cash-in-hand after selling durables grid must exceed that of the liquid assets grid, as households are expected to allocate some resources toward consumption. Furthermore, setting the maximum level of cash-in-hand after selling durables equal to the sum of the tops of the cash-in-hand and durable goods stock grids ensures that the grid is sufficiently wide to facilitate comparisons between the adjust and keep cases.

The labor income autoregressive processes for both permanent and fixed-term workers are discretized into five-state Markov processes using Rouwenhorst's method. In addition to the ten employment states, there are two unemployment states — representing unemployed households receiving unemployment benefits or not — bringing the total number of idiosyncratic states to twelve. The economy can be in either a boom or recession (representing high or low employment risk). Furthermore, there are nine aggregate states for wealth shocks, as both securities and real estate price processes are discretized into three-state Markov processes using Rouwenhorst's method.